GARDEN CITY UNIVERSITY
EMPHASIS ON LIFE

Dr. APJ ABDUL KALAM SCHOOL OF ENGINEERING

DEPARTMENT OF ENGINEERING

# COURSE NAME: AI for Engineers (Intel AI)

# COURSE CODE: 10AFYCC25217

## SYLLABUS:

| Course Code | Course Title | Course Type | | L | T | P | C |
|---|---|---|---|---|---|---|---|
| 10AFYCC25217 | AI for Engineers (Intel AI) | Core (Program) | | 2 | 0 | 2 | 3 |
| **Program** | **Pre-Requisites** | **Semester** | **AI tools/ Tools/ Languages** | | | | |
| BTCE, BTIT, BTRE, BECS, BEIS | Basic Knowledge on Artificial Intelligence | II | Python | | | | |

| | |
|---|---|
| **Course Outcomes** | The student should be able to:<br>1. Understand and appreciate the history of AI, path of development and contributions of several techniques<br>2. List and explain types of agents, their working principles and application scenarios<br>3. Formulate problem using search Algorithms and apply them in different scenarios<br>4. Understand, formulate problems in logic-based reasoning approaches and use appropriate knowledge representation methods. |
| **SYLLABUS** | |
| **Unit – I History of AI** | Early developments and milestones Evolution of AI from symbolic AI to machine learning and deep learning, Fundamental Concepts and Definitions: Definitions of AI Differences between AI, Machine Learning, and Deep Learning, Branches of AI: Natural Language Processing (NLP) Robotics Expert Systems Computer Vision Speech Recognition, Applications of AI in Various Fields: Healthcare Finance Education Transportation Entertainment, Ethical and Societal Implications of AI: AI ethics Bias and fairness in AI Privacy concerns, Impact on jobs and economy, AI governance and regulations |
| **Unit – II Intelligent Agents** | Concept of Intelligent Agents: ` Definition and characteristics of agents, Types of Agents: Simple reflex agents Model-based agents Goal-based agents Utility-based agents, Agent Environments: Fully vs. partially observable Deterministic vs. stochastic Static vs. dynamic Discrete vs. continuous Rationality and Performance Measures: Rational agent definition Performance measures and evaluation Examples of Intelligent Agents: Autonomous robots (e.g., Roomba) Software agents (e.g., chatbots, recommendation systems) Self-driving cars |
| **Unit – III Problem Solving** | Problem-solving Agents and Problem Formulation: Structure of problem-solving agents Formulating problems in AI Search Algorithms: Uninformed search: Breadth-First Search (BFS) Depth-First Search (DFS) Uniform Cost Search Informed search: Greedy Best-First Search A* Search ` Constraint Satisfaction Problems (CSP): Definition and examples of CSP Backtracking search for CSP Local search algorithms Adversarial Search and Game Playing: Minimax algorithm Alpha-beta pruning Applications to games like chess and tic-tac-toe |
| **Unit – IV Knowledge Representation** | Logical Agents and Their Importance: Role of logical agents in AI Propositional Logic: Syntax and semantics Logical connectives and truth tables Inference in propositional logic First-order Logic: Syntax and semantics Quantifiers and predicates Translating natural language to first-order logic Inference in First-order Logic: Forward chaining Backward chaining Resolution Knowledge Representation Issues: Frame problem Qualification problem Common sense knowledge and reasoning |
| **Textbooks** | 1.Intel AI (Introduction to Artificial Intelligence) .Instructor led Intel AI Course.<br>2.S. Russell and P. Norvig, *Artificial Intelligence: A Modern Approach*, 4th ed. Hoboken, NJ, USA: Pearson, 2021.<br>3.S. Haykin, *Neural Networks and Learning Machines*, 3rd ed. Upper Saddle River, NJ, USA: Pearson, 2009. |

# Unit -I: History of AI

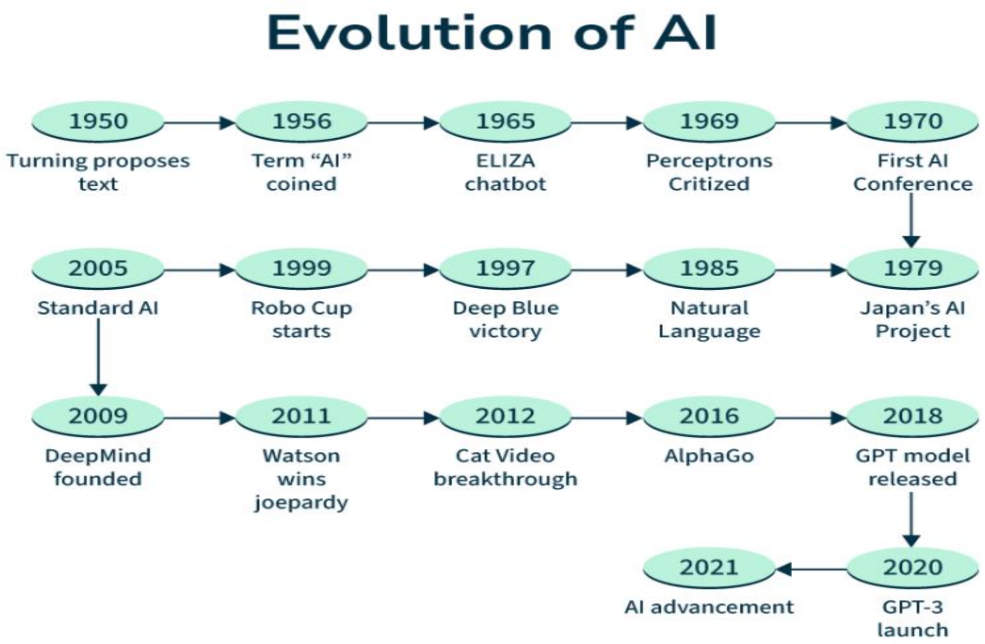
## INTRODUCTION:

Artificial Intelligence (AI) is a branch of computer science that enables machines to simulate human intelligence, allowing them to learn, reason, solve problems, and make decisions. It encompasses technologies like machine learning and neural networks to process data, recognize patterns, and automate tasks in fields such as healthcare, finance, and transportation.

## EARLY DEVELOPMENTS AND MILESTONES

### The Evolution of AI: Major Milestones

The history of AI is often characterized by cycles of great optimism followed by periods of reduced funding and interest, known as "AI Winters."




## The Technical Evolution of AI: From Symbolic AI to machine learning and Deep learning (Logic to Connectionism)

The evolution of Artificial Intelligence is defined by a fundamental shift in how machines "know" things: moving from human-coded logic to data-driven intuition.

## 1. Symbolic AI: The Top-Down Approach (1950s – 1980s)

Also known as **"Good Old-Fashioned AI" (GOFAI)**, this era was based on the "Physical Symbol System Hypothesis."

- **Mechanism:** It relied on **High-Level Symbols** (words, numbers) and **Explicit Rules**. It was a "Top-Down" approach where humans programmed every logical step.
- **Key Technology: Expert Systems.** These programs utilized an "Inference Engine" to process a "Knowledge Base" of if-then rules.
- **The Paradigm:** Intelligence = Logic. If you could define the rules of a problem perfectly, the machine could solve it.
- **The Failure (The "Brittle" Problem):** Symbolic AI could not handle uncertainty, "fuzzy" data, or common-sense reasoning. If a situation fell outside its programmed rules, the system crashed.

## 2. Machine Learning (ML): The Statistical Shift (1990s – 2010s)

As computational power grew and the internet began generating vast amounts of data, the field shifted to a "Bottom-Up" approach.

- **Mechanism:** Instead of being told the rules, the machine uses **Statistical Algorithms** to find patterns in data.
- **Feature Engineering:** In this stage, humans still played a major role by "extracting features." For example, to identify a car, a human would tell the computer to look for wheels, windows, and headlights.
- **Key Algorithms:** Linear Regression, Random Forests, and Support Vector Machines (SVMs).
- **The Paradigm:** Intelligence = Pattern Recognition. The goal is to minimize an "error function" through iterative training on labeled datasets.

## 3. Deep Learning (DL): The Connectionist Revolution (2012 – Present)

Deep Learning is a specialized subset of Machine Learning that utilizes **Artificial Neural Networks (ANNs)** with many hidden layers.

- **Mechanism:** It mimics the structure of the human brain. Unlike traditional ML, Deep Learning performs **Automated Feature Extraction**. The system decides for itself which parts of the data (e.g., specific pixels in an image) are important for the final classification.
- **The Catalyst:** The convergence of **Big Data**, **Massive Parallel Processing** (via GPUs), and algorithmic breakthroughs (like Backpropagation).

- **Key Architectures: * CNNs (Convolutional Neural Networks):** For vision and images.
    - **Transformers:** For language models (like the one powering this conversation).
- **The Paradigm:** Intelligence = Representation Learning. The machine creates a multi-layered, abstract mathematical representation of the world.

## DEFINITION OF ARTIFICIAL INTELLIGENCE

Artificial Intelligence is the branch of computer science that focuses on creating systems capable of performing tasks that normally require human intelligence, such as learning, reasoning, problem-solving, perception, and decision-making.

- Mimics human intelligence
- Broad field covering many techniques

**Examples**

**Virtual Assistants** – Siri

**Chatbots** – Customer support bots on websites

**Expert Systems** – Medical diagnosis systems (e.g., MYCIN)

**Game Playing Systems** – IBM Deep Blue

**Autonomous Robots** – Cleaning robots like Roomba

### DEFINITION OF MACHINE LANGUAGE

Machine Learning is a subset of Artificial Intelligence that enables systems to learn from data and improve their performance without being explicitly programmed.

- Learns from data
- Improves automatically with experience

**Examples:**

- **Email Spam Filtering**
  → Classifies emails as spam or not spam based on past data
- **Recommendation Systems** – Netflix, Amazon
  → Suggest movies or products based on user behavior
- **Fraud Detection in Banking**
  → Identifies unusual transactions
- **Predictive Text / Auto-complete**
  → Suggests next words while typing

## DEFINITION OF DEEP LEARNING

Deep Learning is a subset of Machine Learning that uses artificial neural networks with multiple layers to model complex patterns in large amounts of data.

- Uses neural networks
- Requires large datasets and high computational power

**Examples:**

- **Image Recognition**
  → Identifying objects in photos (face recognition in phones)
- **Speech Recognition** – Google Assistant
  → Converts speech to text
- **Self-Driving Cars** – Tesla
  → Uses deep neural networks for navigation
- **AI in Games** – AlphaGo
  → Defeated world champion in Go using deep learning

### KEY POINTS

**AI Examples are based on:** Rule-based system or assistant

**ML Examples are based on:** Recommendation or prediction

**DL Examples are based on:** Image, speech, or self-driving

## DIFFERENCE BETWEEN AI, ML AND DL

| Aspect | Artificial Intelligence (AI) | Machine Learning (ML) | Deep Learning (DL) |
|---|---|---|---|
| **Definition** | Broad field of creating intelligent machines | Subset of AI that learns from data | Subset of ML using neural networks |
| **Goal** | Simulate human intelligence | Enable machines to learn automatically | Mimic human brain using deep neural networks |
| **Scope** | Wide (includes ML, DL, robotics, etc.) | Narrower than AI | Narrowest (subset of ML) |
| **Approach** | Rule-based + learning | Data-driven algorithms | Multi-layer neural networks |
| **Human Intervention** | High | Moderate | Low |
| **Data Requirement** | Low to Medium | High | Very High |
| **Complexity** | Less complex | Moderate | Highly complex |
| **Performance** | Limited in complex tasks | Good performance | Excellent in complex tasks |
| **Examples** | Siri, Expert systems | Spam filtering, Netflix recommendations | Image recognition, AlphaGo |

## Branches of Artificial Intelligence

Artificial Intelligence is a vast field divided into several specialized branches, each focusing on a specific aspect of mimicking human-like capabilities.

## 1. Natural Language Processing (NLP)

NLP is the branch of AI that enables computers to understand, interpret, and generate human language (text and speech).

- **Key Tasks:** Machine translation (e.g., Google Translate), sentiment analysis, text summarization, and chatbots.
- **Mechanism:** It combines computational linguistics with statistical, machine learning, and deep learning models to process "natural" human language which is often ambiguous and complex.

## 2. Computer Vision

Computer Vision allows machines to "see" and interpret visual information from the world, such as images and videos.

- **Key Tasks:** Object detection, facial recognition, image classification, and medical image analysis (e.g., detecting tumors in X-rays).
- **Mechanism:** It primarily uses **Convolutional Neural Networks (CNNs)** to identify patterns, edges, and textures in digital images to reconstruct a 3D understanding of a 2D input.

## 3. Robotics

Robotics is the intersection of AI, electrical engineering, and mechanical engineering. It involves creating machines (robots) that can perform physical tasks.

- **Key Tasks:** Industrial automation, autonomous drones, surgical robots, and space exploration (e.g., Mars rovers).
- **Mechanism:** AI provides the "brain" for the robot, allowing it to use sensors to perceive its environment and planning algorithms to navigate or manipulate objects safely.

## 4. Expert Systems

One of the earliest branches of AI, Expert Systems are designed to solve complex problems by reasoning through bodies of knowledge, mimicking the decision-making ability of a human expert.

- **Key Tasks:** Medical diagnosis, credit scoring, and chemical analysis.

- **Structure:**
    1. **Knowledge Base:** A repository of facts and rules provided by human experts.
    2. **Inference Engine:** The "logical" part that applies the rules to the known facts to deduce new information.

## 5. Speech Recognition

While often grouped with NLP, Speech Recognition specifically focuses on converting spoken language into text.

- **Key Tasks:** Voice-to-text dictation, voice-controlled UI (like Alexa or Siri), and automated phone systems.
- **Mechanism:** It involves acoustic modelling (understanding phonemes/sounds) and language modelling (understanding the probability of certain words following others) to accurately transcribe speech despite different accents and background noise.

## Applications of AI in Various Fields:

## 1. Healthcare

AI is transforming medicine from a reactive practice to a proactive, personalized science.

- **Medical Imaging:** AI algorithms analyze X-rays, MRIs, and CT scans to detect anomalies like tumors or fractures with accuracy often surpassing human specialists.
- **Drug Discovery:** Machine learning models simulate molecular interactions, reducing the time and cost of bringing new life-saving drugs to market.
- **Predictive Analytics:** Systems monitor patient vitals to predict potential health crises, such as sepsis or cardiac arrest, before they occur.

## 2. Finance

In the financial sector, AI is primarily used for risk management and the automation of complex data processing.

- **Fraud Detection:** AI monitors millions of transactions in real-time, identifying patterns that deviate from a user's typical behavior to flag potential credit card fraud.
- **Algorithmic Trading:** High-frequency trading platforms use AI to execute trades at speeds and volumes impossible for humans, reacting to market changes in milliseconds.
- **Credit Scoring:** Beyond traditional credit history, AI evaluates alternative data points to provide more accurate risk assessments for loans.

## 3. Education

AI facilitates a shift toward "student-centric" learning models.

- **Personalized Learning:** Adaptive platforms analyze a student's performance in real-time, adjusting the difficulty and type of content to match their specific learning pace.
- **Administrative Automation:** AI tools assist educators by automating grading for multiple-choice and short-answer assessments, allowing teachers to focus more on mentorship.
- **Intelligent Tutoring:** AI-powered bots provide 24/7 support to students, answering common questions and guiding them through complex problem-solving steps.

## 4. Transportation

The focus here is on increasing safety and optimizing the movement of goods and people.

- **Autonomous Vehicles:** Self-driving cars and trucks use computer vision and sensor fusion to navigate traffic, recognize pedestrians, and make split-second safety decisions.
- **Traffic Management:** Cities use AI to analyze traffic flow and adjust signal timing dynamically, significantly reducing congestion and carbon emissions.
- **Logistics Optimization:** AI predicts the most efficient routes for delivery fleets, accounting for weather, road conditions, and fuel consumption.

## 5. Entertainment

AI enhances both the creation and the consumption of digital content.

- **Recommendation Engines:** Streaming services like Netflix and Spotify use collaborative filtering to suggest content based on your viewing or listening history.
- **Content Generation:** AI is used to render realistic visual effects (VFX) in movies and to generate procedural environments in video games.
- **Deepfake & Synthesis:** Advanced AI can synthesize realistic human speech or alter video footage, used both for dubbing movies and creating virtual influencers.

## Ethical and Societal Implications of AI

Artificial Intelligence (AI) is transforming industries, economies, and daily life. However, its rapid growth raises important **ethical and societal concerns** that must be addressed to ensure responsible use.

AI Ethics refers to the moral principles and guidelines that govern the design, development, and use of artificial intelligence systems.

## Key Principles of AI Ethics

- **Transparency** – AI decisions should be explainable and understandable
- **Accountability** – Developers and organizations are responsible for AI outcomes
- **Fairness** – AI should not discriminate against individuals or groups
- **Privacy** – Protection of user data and personal information
- **Safety & Security** – AI systems should be reliable and secure
- **Human Control** – Humans should remain in control of AI systems

## Importance

- Prevents misuse of AI
- Builds trust among users
- Ensures responsible innovation

## AI ethics Bias and fairness in AI Privacy concerns

### 1. AI Ethics: The "Rules of the Road"

Ethics in AI is about ensuring that technology helps society without causing accidental harm. Think of it as a code of conduct for machines.

- **Accountability**: If an AI makes a mistake (like a self-driving car crash), who is responsible? The programmer, the owner, or the company?
- **Transparency**: An AI should be able to explain *why* it made a decision. This is called "Explainable AI."
- **Human Oversight**: AI should assist humans, not replace human judgment in critical areas like law or medicine. This is often called having a "Human-in-the-Loop."

## 2. Bias and Fairness: The "Mirror" Problem

AI isn't born with "opinions." It learns by looking at data from the past. If that data is unfair, the AI will be unfair too. This is known as **"Garbage In, Garbage Out."**

- **What is Bias?** It occurs when an AI system produces results that are systematically prejudiced against certain groups of people (based on race, gender, age, etc.).
- **How it happens:**
    - **Data Bias:** If a medical AI is only trained on data from men, it might not work well for women.

- **Historical Bias:** If a hiring AI looks at past successful managers and sees mostly men, it might incorrectly learn that "being a man" is a requirement for the job.

- **The Goal of Fairness:** Ensuring that the AI treats every individual or group equally, regardless of their background.

## 3. Privacy: Protecting Your "Digital Twin"

AI needs massive amounts of data to work. The concern is how that data is collected, stored, and used.

- **Data Collection:** Many AI systems "scrape" the internet for information. Did the people who created that data give permission for it to be used to train an AI?
- **Surveillance:** AI-powered facial recognition can track people in real-time. This raises concerns about the "right to be anonymous" in public.
- **Inference:** AI is so smart that it can guess private things about you (like your health status or political views) just by looking at your "likes" or "buying habits," even if you never told the AI that information.

## AI Impact on jobs and economy

AI is changing the way we earn money and work.

- **Automation:** AI handles "boring" tasks (like filing data), which might cut some simple office jobs.
- **New Skills:** To stay hired, people need to learn how to **work with AI** (Reskilling).
- **Better Productivity:** AI helps businesses work faster, which can make the global economy grow.

| Issue | Problem | Solution |
|---|---|---|
| **Ethics** | AI doing "wrong" things. | Set strict rules and laws. |
| **Bias** | AI being "unfair." | Use diverse, honest data. |
| **Privacy** | Your data being stolen. | Stronger passwords and privacy laws. |
| **Jobs** | Robots taking jobs. | Humans learning new tech skills. |

## AI governance and regulations

**What is AI Governance?**

It is the set of **rules and laws** that tell companies and governments how they are allowed to build and use AI. The goal is to make sure AI is a "helper" and not a "threat."

## <u>Objectives of AI Governance</u>

- Ensure **ethical use of AI**
- Protect **human rights and privacy**
- Promote **fairness and non-discrimination**
- Maintain **transparency and accountability**
- Reduce **risks and misuse of AI systems.**

## The "Risk" Categories

Most countries (like those in the EU) now group AI into levels of risk:

- **Unacceptable Risk (Banned):** AI that tracks your behaviour to give you a "Social Score" or manipulates your mind.
- **High Risk (Strictly Regulated):** AI used in **Hospitals, Schools, or Hiring.** These must be checked by humans and have an "off switch."
- **Limited Risk:** AI like Chatbots. These just need to be **honest** (e.g., they must tell you "I am an AI").

## Key Global Rules (2026)

- **Transparency:** If an AI makes a picture or video, it must have a **label** (like a digital watermark) so people know it isn't real.
- **Explainability:** If an AI rejects your loan or job application, the company **must explain why** in simple words.
- **Privacy:** AI cannot use your private data without clear permission.

| Rule | Meaning |
|---|---|
| **Accountability** | A human is always responsible for the AI's actions. |
| **Safety** | AI must be tested to ensure it doesn't "break" or get hacked. |
| **Fairness** | Laws require companies to prove their AI isn't biased. |
| **Labels** | AI-generated content must be clearly marked. |

**END**